

Concept: Linear equations in one unknown

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Linear equations in one unknown: Source questions

NCERT Exemplar · 2024 · Class 8/9 Mathematics · QQ6 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept: A linear equation in one variable is true only for its solutions: test a value by evaluating the LHS and the RHS

Concept difficulty: Easy concept

Question difficulty: Connected reasoning

Family: Deciding and finding solutions by substitution

Question type: Single-correct multiple choice

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

The value of x for which the expressions $3x - 4$ and $2x + 1$ become equal is

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Conditions

- x is one variable; the expressions are equal at x when they take the same value there.
- Exactly one option is correct.

(A) (a) -3

(B) (b) 0

(C) (c) 5

(D) (d) 1

NCERT Exemplar · 2024 · Class 8/9 Mathematics · QQ17 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept: A linear equation in one variable is true only for its solutions: test a value by evaluating the LHS and the RHS

Concept difficulty: Easy concept

Question difficulty: Connected reasoning

Family: Deciding and finding solutions by substitution

Question type: Constructed response

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

The solution of the equation $3x - 4 = 1 - 2x$ is _____.

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Conditions

- The equation is linear in the one variable x .
- A solution makes the LHS and RHS equal.

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Subject: Mathematics

Topic: Linear equations

Concept: A linear equation in one variable is true only for its solutions: test a value by evaluating the LHS and the RHS

Concept difficulty: Easy concept

Question difficulty: Connected reasoning

Family: Deciding and finding solutions by substitution

Question type: Constructed response

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

$9x - \underline{\hspace{2cm}} = -21$ has the solution (-2)

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Conditions

- The blank holds one number.
- $x = -2$ must make the two sides equal.

NCERT Textbook · 2024 · Class 8/9 Mathematics · QQ1 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept:

Solving an equation with the variable on both sides: the same operation on both sides, and transposing a variable term like a number

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Equations with the variable on both sides

Question type: Constructed response

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

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Solve the following equation and check
your result: $3x = 2x + 18$

Conditions

- Solve the equation and check the result, as the exercise asks.
- Adding or subtracting the same number on both sides, or multiplying or dividing both sides by the same non-zero number, does not change the solution.

NCERT Textbook · 2024 · Class 8/9 Mathematics · QQ2 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept:

Solving an equation with the variable on both sides: the same operation on both sides, and transposing a variable term like a number

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Equations with the variable on both sides

Question type: Constructed response

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

Solve the following equation and check your result: $5t - 3 = 3t - 5$

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Conditions

- Solve the equation and check the result, as the exercise asks.
- Adding or subtracting the same number on both sides, or multiplying or dividing both sides by the same non-zero number, does not change the solution.

NCERT Textbook · 2024 · Class 8/9 Mathematics · QQ3 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept:

Solving an equation with the variable on both sides: the same operation on both sides, and transposing a variable term like a number

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Equations with the variable on both sides

Question type: Constructed response

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

Solve the following equation and check your result: $5x + 9 = 5 + 3x$

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Conditions

- Solve the equation and check the result, as the exercise asks.
- Adding or subtracting the same number on both sides, or multiplying or dividing both sides by the same non-zero number, does not change the solution.

NCERT Textbook · 2024 · Class 8/9 Mathematics · QQ5 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept:

Solving an equation with the variable on both sides: the same operation on both sides, and transposing a variable term like a number

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Equations with the variable on both sides

Question type: Constructed response

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

Solve the following equation and check your result: $2x - 1 = 14 - x$

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Conditions

- Solve the equation and check the result, as the exercise asks.
- Adding or subtracting the same number on both sides, or multiplying or dividing

both sides by the same non-zero number,
does not change the solution.

NCERT Exemplar · 2024 · Class 8/9 Mathematics · QQ16 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept:

Solving an equation with the variable on both sides: the same operation on both sides, and transposing a variable term like a number

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Equations with the variable on both sides

Question type: Single-correct multiple choice

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

If we multiply or divide both sides of a linear equation with a non-zero number, then the solution of the linear equation :

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Conditions

- The multiplier or divisor is the same non-zero number on both sides.

(A) (A) Changes

(B) (B) Remains the same

(C) (C) Changes in case of multiplication only

(D) (D) Changes in case of division only

NCERT Textbook · 2024 · Class 8/9 Mathematics · QQ7 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept: Reducing an equation to simpler form: opening brackets, combining like terms and clearing denominators before solving

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Reducing equations to simpler form

Question type: Constructed response

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

Simplify and solve the following linear equation: $3(t - 3) = 5(2t + 1)$

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Conditions

- Open every bracket and combine like terms on each side before collecting terms across the equality sign.
- The same operation on both whole sides (multiplying or dividing only by a non-zero number) keeps the solution.

NCERT Textbook · 2024 · Class 8/9 Mathematics · QQ9 Source unverified

Subject: Mathematics

Topic: Linear equations

Concept: Reducing an equation to simpler form: opening brackets, combining like terms and clearing denominators before solving

Concept difficulty: Medium concept

Question difficulty: Multi-step reasoning

Family: Reducing equations to simpler form

Question type: Constructed response

Source: Legacy textbook/exemplar wording; verification pending

Provenance: Source unverified

ATTEMPT FIRST

Simplify and solve the following linear equation: $3(5z - 7) - 2(9z - 11) = 4(8z - 13) - 17$

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Conditions

- Open every bracket and combine like terms on each side before collecting terms across the equality sign.
- The same operation on both whole sides (multiplying or dividing only by a non-zero number) keeps the solution.